The B-S model formula for put and call option prices is as follows:

$$c = S_0\, N(d_1) - K\, e^{-rT} N(d_2)$$
$$p = K\, e^{-rT}\, N(-d_2) - S_0\, N(-d_1)$$
$$\text{where} \quad d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$
$$d_2 = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

C = Call option price
P = Put option price
N = Normal distribution
S_o = Spot price of an asset
K = Strike price
R = Risk-free interest rate
t = The time to maturity
σ = The volatility of the asset

Image Source: https://www.youtube.com/watch?v=3fWJ8DuT4fQ

This formula may appear daunting at first. However, you can use various online B-S option price calculators. So, as long as you understand its assumptions, limitations, and usefulness, you can derive valuable insights from the B-S model.

This model makes the following assumptions:

- No dividends are paid out during the life of the option.
- Markets are random because one cannot predict market movements.
- No transaction costs are involved in buying.
- There's no change in the risk-free rate and volatility of the underlying asset. These are known constants.
- The underlying asset returns follow a normal distribution.
- The formula assumes European options.
- There are no arbitrage opportunities, i.e., the market instantly updates the stock price for material events.
- Prices are lognormally distributed. So, they follow a statistical distribution of logarithmic values from the corresponding normal distribution.